

# Functional and differential equations

## — 3

Qualitative theory: answers without solving the equation

13 сентября 2026 года

### Theoretical minimum.

**Theorem 1 (existence, uniqueness, continuation).** For  $y' = f(x, y)$  with  $f$  continuous a solution exists locally (Peano) and is unique if  $f$  is Lipschitz in  $y$  (Picard, proved through the contraction principle in Functional sequences, series and spaces – 4). A solution extends until it leaves every compact set: it can only stop existing by blowing up or by reaching the boundary of the domain.

**Theorem 2 (the phase line and the phase plane).** For an autonomous equation  $\dot{x} = f(x)$  on  $\mathbb{R}$  the zeros of  $f$  split the line into intervals of monotone motion, so every bounded solution tends to an equilibrium and periodic solutions are impossible. In the plane the picture is richer, but a trajectory cannot cross itself, which already forbids many configurations.

**Theorem 3 (Lyapunov functions).** If  $V \geq 0$  vanishes only at the equilibrium and  $\frac{d}{dt}V(x(t)) \leq 0$  along the trajectories, the equilibrium is stable; if the derivative is strictly negative away from it, it is asymptotically stable. A first integral — a function constant along the trajectories — confines the motion to a level set, which for a conservative system replaces the solution entirely.

**Fact 1 (what to ask instead of a formula).** Is the solution defined on the whole line? Is it bounded? Does it tend to a limit? Does it oscillate, and how far apart are the zeros (Differentiable functions – 6)? Does a trajectory return? Almost every olympiad problem about a differential equation asks one of these five questions, and none of them requires solving it.

- 
1. Prove that a solution of  $y' = f(x, y)$  with continuous  $f$  extends until it leaves every compact subset of the domain; construct an equation whose solution blows up in finite time and one whose solutions are non-unique, and prove that  $y' = \sqrt{|y|}$ ,  $y(0) = 0$  has infinitely many solutions.
  2. (Kapacëв 4.1) Suppose the equation  $y' = f(x, y)$  has a unique solution  $y(x)$ , defined on the whole line, for every initial condition  $y(x_0) = y_0$ . Does it follow that  $f$  must be continuous?
  3. (Kapacëв 4.2) Prove that the equation  $x^3 y'' + y = 0$  on  $(0, +\infty)$  has a solution tending to  $+\infty$  as  $x \rightarrow +\infty$ .
  4. (Kapacëв 4.11) Let the right-hand side of an autonomous system  $\dot{x} = f(x)$  in  $\mathbb{R}^n$  be continuously differentiable and satisfy  $|f(x_1) - f(x_2)| \leq L|x_1 - x_2|$ . Prove that if a trajectory  $x(t)$  is non-constant and periodic with period  $T$ , then  $T \geq \frac{2\pi}{L}$ .
  5. For the system  $\dot{x} = y$ ,  $\dot{y} = -\sin x$  find a first integral, draw the phase portrait, prove that the equilibrium at the origin is stable but not asymptotically stable, and determine which trajectories are periodic.